

ECON 636 600: MACROECONOMIC THEORY I

Assignment 4 (due on 10:00am Tuesday, Nov 22)

1. (20 points) Suppose $\{x_t; t \geq 0\}$ is a markov chain with transition matrix $P_{n \times n}$. A **stationary** (**or invariant**) distribution for P is a probability vector such that $\pi = \pi P$. Find out the invariant distribution for

(a) $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$

$\pi = (4/7, 3/7)$

(b) $P = \begin{pmatrix} 1 & 0 \\ 0.9 & 0.1 \end{pmatrix}$

$\pi = (1, 0)$

2. (80 points) (Macro qualifier at Stanford University) A shopper wants to purchase one unit of an indivisible consumption good at the lowest possible cost. There are multiple stores that she can visit. First, there is a small neighborhood convenience store. In addition, there is a mall with an infinite number of stores in it. Each of these stores offers the good at a random price. For each store, the price is drawn independently from the same uniform distribution over the interval $[0, B]$.

The agent can follow one of two strategies. Either she can go immediately to the mall and start searching there (strategy A), or she can first check the convenience store and continue to the mall only if she is not satisfied with the local price (strategy B).

The cost of traveling to the convenience store and examining the price there is c_1 . Once at the convenience store, the agent holding the item with the price tag p has to choices. She can either purchase the item and stop shopping, or she can leave the item and continue to the

mall.

The cost of traveling to the mall either from home or from the convenience store is $c_2 > c_1$. (This includes the cost of checking the price in the first store in the mall.) Again, the shopper can either accept this first price and stop shopping or she can reject it and go to the next store in the mall. The cost of going to the next store and checking the price there is $\epsilon < c_1$. In the mall, she can continue shopping as long as she wishes. Visiting each new store costs an additional amount ϵ . Assume that $\epsilon < \frac{B}{2}$.

The agent aims to **minimize** the total cost of shopping, which is the sum of travel costs and the price that is ultimately paid. You can assume that there is no recall of past prices. Note also that there is no discounting.

- (a) Let $v_2(p)$ denote the value of expected future costs for the shopper if she is in one of the stores in the mall holding an item priced p . Describe her Bellman equation. Show that her optimal strategy is to follow a reservation price strategy, i.e., buy the item if $p \leq \bar{p}_2$, otherwise continue shopping.

$$v_2(p) = \min\{p, \epsilon + \int_0^B v_2(p') \frac{1}{B} dp'\}$$

The solution is clearly a reservation price strategy.

- (b) Show that the reservation price at the mall satisfies

$$\bar{p}_2 = \sqrt{2B\epsilon}.$$

the condition for the reservation price is

$$\epsilon = \int_0^{\bar{p}_2} (p' - \bar{p}_2) \frac{1}{B} dp',$$

or

$$\epsilon = \frac{(\bar{p}_2)^2}{2B}.$$

- (c) Let $v_1(p)$ denote the value of expected future costs for the shopper if she is in the convenience store holding an item priced p . Show that her optimal strategy is also a reservation price strategy, i.e., buy the item if $p \leq \bar{p}_1$, otherwise continue shopping.

$$v_1(p) = \min\{p, \quad c_2 + \int_0^B v_2(p') \frac{1}{B} dp'\}$$

The solution is clearly a reservation price strategy.

- (d) Show that the reservation price at the convenience store satisfies

$$\bar{p}_1 = c_2 - \epsilon + \sqrt{2B\epsilon}.$$

the condition for the reservation price is

$$\bar{p}_1 = c_2 + \int_0^B v_2(p') \frac{1}{B} dp',$$

or

$$\bar{p}_1 = c_2 + \bar{p}_2 - \epsilon.$$

- (e) Let v^A denote the expected cost of the shopper if she follows strategy A , and v^B denote her expected cost if she follows strategy B . Find analytical expressions for v^A and v^B .

$$v^A = c_1 + \int_0^B v_1(p) \frac{1}{B} dp = c_1 + \frac{\bar{p}_1^2}{2B} + \frac{B - \bar{p}_1}{B} \bar{p}_1 = c_1 - \frac{\bar{p}_1^2}{2B} + \bar{p}_1.$$

Similarly

$$v^B = c_2 - \frac{\bar{p}_2^2}{2B} + \bar{p}_2.$$

- (f) Under what conditions will the shopper prefer to check the convenience store before heading to the mall?

$$c_1 - \frac{\bar{p}_1^2}{2B} + \bar{p}_1 \leq c_2 - \frac{\bar{p}_2^2}{2B} + \bar{p}_2.$$